Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	1
Number of Questions to be attempted :	1
Section Marks :	25
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065355414
Question Shuffling Allowed :	Yes

Question Number : 1 Question Id : 640653386919 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 25

Question Label : Multiple Choice Question

THERE IS NO SPEECH TECHNOLOGY QP IN FORENOON SESSION

Options :

6406531286714. ✔ YES

6406531286715. ✖ NO

Industry 4.0

Section Id :	64065323913
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory

Number of Questions :	10
Number of Questions to be attempted :	10
Section Marks :	20
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065355415
Question Shuffling Allowed :	No

Question Number : 2 Question Id : 640653386920 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL: INDUSTRY 4.0"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531286716. ✔ YES

6406531286717. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065355416
Question Shuffling Allowed :	Yes

**Question Number : 3 Question Id : 640653386921 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the correct progression of the industry?

Options :

6406531286718. ✖ Steam engine -> IT ->Assembly line ->IOT

6406531286719. ✔ Steam engine -> Assembly line ->IT ->IOT

6406531286720. ✖ Assembly line ->Steam engine -> IT ->IOT

6406531286721. ✖ Assembly line ->IT -> Steam engine ->IOT

Question Number : 4 Question Id : 640653386922 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which is the country of origin for Industry 4.0?

Options :

6406531286722. ✖ Japan

6406531286723. ✖ United States

6406531286724. ✔ Germany

6406531286725. ✖ United Kingdom

Question Number : 5 Question Id : 640653386923 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the Industry 4.0 version of transportation.

Options :

6406531286726. ✖ IC engine

6406531286727. ✖ Electric car

6406531286728. ✔ Hyperloop

6406531286729. ✖ Steam locomotive

Question Number : 6 Question Id : 640653386930 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Location problem is dependent on which of the following factors?

Options :

6406531286738. ✖ Distance

6406531286739. ✖ Number of facilities

6406531286740. ✖ Optimization criteria

6406531286741. ✔ All of these

Question Number : 7 Question Id : 640653386931 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the below formula represent the Metropolitan metric distance?

Options :

6406531286742. ✖ $d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$

6406531286743. ✔ $d_{ij} = |(x_i - x_j)| + |(y_i - y_j)|$

6406531286744. ✖ $(x2 - x1) + (y2 - y1)$

6406531286745. ✖ All of these

Question Number : 8 Question Id : 640653386932 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

If we choose the **Euclidean** distance measure, how will the optimal solution be?

Options :

6406531286746. ✔ Euclidean distance measure will result in a single point optimal solution

6406531286747. ✖ Metropolitan metric distance will result in a range for the optimal solution

6406531286748. ✖ Both Euclidean distance measure will result in a single point optimal solution & Metropolitan metric distance will result in a range for the optimal solution are wrong

6406531286749. ✖ Both Euclidean distance measure will result in a single point optimal solution & Metropolitan metric distance will result in a range for the optimal solution are correct

Question Number : 9 Question Id : 640653386933 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Choice Question

If we choose **Metropolitan distance** measure, how will the optimal solution be?

Options :

6406531286750. ✖ Euclidean distance measure will result in a single point optimal solution

6406531286751. ✔ Metropolitan metric distance will result in a range for the optimal solution

6406531286752. ✖ Both Euclidean distance measure will result in a single point optimal solution & Metropolitan metric distance will result in a range for the optimal solution are wrong

6406531286753. ✖ Both Euclidean distance measure will result in a single point optimal solution & Metropolitan metric distance will result in a range for the optimal solution are correct

Sub-Section Number : 3

Sub-Section Id : 64065355417

Question Shuffling Allowed : No

Question Id : 640653386924 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (10 to 14)

Question Label : Comprehension

Currently, a company is using a layout that consists of 9 workstations. 200 sandwiches are produced in a day on this layout. A typical day consists of a single shift that runs for 10 hours. The sandwich-making process consists of 9 activities (the information is provided in the table below). Answer the given subquestions .

Activity ID	Activity Description	Activity Time (in seconds)	Preceding Activities
A	Wheat bag emptying	20	
B	Vegetable cleaning	10	
C	Bread baking	60	A
D	Vegetable cutting	40	B
E	Bread Cutting	20	C
F	Butter applying	5	E
G	Vegetable filling	5	F, E
H	Sandwich baking	15	G
I	Sandwich packing	5	H

Sub questions

Question Number : 10 Question Id : 640653386925 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the current efficiency of the line where each activity is performed on a separate workstation (enter only the numerical value up to two decimal places without the “%” symbol. Example: if the answer is 99.756%” enter only “99.76”)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

11.10 to 11.12

Question Number : 11 Question Id : 640653386926 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the maximum possible output that can be achieved?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

600

Question Number : 12 Question Id : 640653386927 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Given the activity information, (theoretically) how many workstations can be reduced from the current line while achieving the maximum possible output?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Question Number : 13 Question Id : 640653386928 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Multiple Select Question

Which among the following layout is the best possible one to achieve the maximum possible output? (choose all that is applicable)

Options :

6406531286733. ✔ (A, B)→(C)→(D, E)→ (F, G, H, I)

6406531286734. ✖ (A)→(C)→(B, E, F)→ (D, G, H) → (I)

6406531286735. ✔ (A, B)→(C)→(D)→ (E, F, G, H, I)

6406531286736. ✖ (A, B)→(C)→(D, G)→ (E, F, H, I)

Question Number : 14 Question Id : 640653386929 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

What is the efficiency of the best possible layout chosen in the previous question (entre only the numerical value up to two decimal places without the “%” symbol. Example: if the answer is 99.756%” enter only “99.76”)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

75

Sub-Section Number : 4

Sub-Section Id : 64065355418

Question Shuffling Allowed : No

Question Id : 640653386934 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (15 to 18)

Question Label : Comprehension

Consider the following table, for the supplier selection problem. This table contains supplier attributes for 10 suppliers as explained in the tutorial. Answer the given subquestions

Supply network	Employee skills	Quality reputation	Price premium	Customers	Management
3	1	3	3	2	1
2	2	1	1	1	2
1	1	1	1	2	1
2	3	2	2	3	3
3	3	3	3	3	2
3	2	2	3	2	3
2	1	3	2	3	2
3	2	3	3	1	3
3	2	3	3	2	1
1	1	1	1	1	1

Sub questions

Question Number : 15 Question Id : 640653386935 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Multiple Choice Question

How many itemsets are there with cardinality/number of elements equal to 3?

Options :

6406531286754. ✖ 3060

6406531286755. ✖ 8538

6406531286756. ✔ 816

6406531286757. ✖ 18

Question Number : 16 Question Id : 640653386936 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1.5

Question Label : Multiple Choice Question

How many itemsets have support value greater than or equal to 0.3?

Options :

6406531286758. ✔ 13

6406531286759. ✖ 9

6406531286760. ✖ 21

6406531286761. ✖ 7

Question Number : 17 Question Id : 640653386937 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

What is the probability of observing {Price_Premium = 3, Quality_reputation =3}?

Options :

6406531286762. ✖ 0.33

6406531286763. ✖ 0.55

6406531286764. ✔ 0.4

6406531286765. ✖ 0.3

Question Number : 18 Question Id : 640653386938 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction